

Unit – V

Risk & Configuration Management

What is Risk?

"Tomorrow problems are today's risk." Hence, a clear definition of a "risk" is a problem that could cause some loss or threaten the progress of the project, but which has not happened yet.

These potential issues might harm cost, schedule or technical success of the project and the quality of our software device, or project team morale.

Risk Management

Risk Management is the system of identifying addressing and eliminating these problems before they can damage the project.

OR

A software project can be concerned with a large variety of risks. In order to be adept to systematically identify the significant risks which might affect a software project, it is essential to classify risks into different classes. The project manager can then check which risks from each class are relevant to the project.



3. Risk Management Practices:

- Software Risk Evaluation (SRE)
- Continuous Risk Management (CRM)
- Team Risk Management (TRM)

There are three main classifications of risks which can affect a software project:

1. Project risks
2. Technical risks
3. Business risks

1. Project risks: Project risks concern different forms of budgetary, schedule, personnel, resource, and customer-related problems. A vital project risk is schedule slippage. Since the software is intangible, it is very tough to monitor and control a software project. It is very tough to control something which cannot be identified. For any manufacturing program, such as the manufacturing of cars, the plan executive can recognize the product taking shape.

2. Technical risks: Technical risks concern potential method, implementation, interfacing, testing, and maintenance issue. It also consists of an ambiguous specification, incomplete specification, changing specification, technical uncertainty, and technical obsolescence. Most technical risks appear due to the development team's insufficient knowledge about the project.

3. Business risks: This type of risks contain risks of building an excellent product that no one needs, losing budgetary or personnel commitments, etc.

Other risk categories

1. Known risks: Those risks that can be uncovered after careful assessment of the project program, the business and technical environment in which the plan is being developed, and more reliable data sources (e.g., unrealistic delivery date)

2. Predictable risks: Those risks that are hypothesized from previous project experience (e.g., past turnover)

3. Unpredictable risks: Those risks that can and do occur, but are extremely tough to identify in advance.

Project Risks:

- Supplier issues
- Organizational factors
- Technical issues

Product Risks :

Below are some of the product risks occurring in a LIVE environment:

- Defect Prone Software delivered
- The Critical defects in the product that could cause harm to an individual (injury or death) or company
- Poor software Features
- Inconsistent Software Features

OR

Various Kinds of Risks in Software Development :

1. **Schedule Risk :**

Schedule related risks refers to time related risks or project delivery related planning risks. The wrong schedule affects the project development and delivery. These risks are mainly indicates to running behind time as a result project development doesn't progress timely and it directly impacts to delivery of project. Finally if schedule risks are not managed properly it gives rise to project failure and at last it affect to organization/company economy very badly.

Some reasons for Schedule risks –

- Time is not estimated perfectly
- Improper resource allocation
- Tracking of resources like system, skill, staff etc
- Frequent project scope expansion
- Failure in function identification and its' completion

2. **Budget Risk:**

Budget related risks refers to the monetary risks mainly it occurs due to budget overruns. Always the financial aspect for the project should be managed as per decided but if financial aspect of project mismanaged then there budget concerns will arise by giving rise to budget risks. So proper

finance distribution and management are required for the success of project otherwise it may lead to project failure.

Some reasons for Budget risks –

- Wrong/Improper budget estimation
- Unexpected Project Scope expansion
- Mismanagement in budget handling
- Cost overruns
- Improper tracking of Budget

3. **Operational Risks:**

Operational risk refers to the procedural risks means these are the risks which happen in day-to-day operational activities during project development due to improper process implementation or some external operational risks.

Some reasons for Operational risks –

- Insufficient resources
- Conflict between tasks and employees
- Improper management of tasks
- No proper planning about project
- Less number of skilled people
- Lack of communication and cooperation
- Lack of clarity in roles and responsibilities
- Insufficient training

4. **Technical Risks :**

Technical risks refers to the functional risk or performance risk which means this technical risk mainly associated with functionality of product or performance part of the software product.

Some reasons for Technical risks –

- Frequent changes in requirement
- Less use of future technologies
- Less number of skilled employee
- High complexity in implementation
- Improper integration of modules

5. **Programmatic Risks:**

Programmatic risks refers to the external risk or other unavoidable risks. These are the external risks which are unavoidable in nature. These risks come from outside and it is out of control of programs.

Some reasons for Programmatic risks –

1. Rapid development of market
2. Running out of fund / Limited fund for project development
3. Changes in Government rules/policy
4. Loss of contracts due to any reason

Risk Identification

It is the procedure of determining which risk may affect the project most. This process involves documentation of existing risks.

OR

Risk identification is the process of determining risks that could potentially prevent the program, enterprise, or investment from achieving its objectives. It includes documenting and communicating the concern.

The risk Identification process itself follows a defined structure and is elaborated progressively through six stages:

1. Template specification
2. Basic identification
3. Detailed identification
4. External cross-check
5. Internal cross-check
6. Statement finalization

Risk Projection

Risk projection, also called risk estimation, attempts to rate each risk in two ways—the likelihood or probability that the risk is real and the consequences of the problems associated with the risk, should it occur.

The project planner, along with other managers and technical staff, performs four risk projection activities:

- (1) establish a scale that reflects the perceived likelihood of a risk,
- (2) delineate the consequences of the risk,
- (3) estimate the impact of the risk on the project and the product, and
- (4) note the overall accuracy of the risk projection so that there will be no misunderstandings.

Risk Refinement

Risk Refinement. **Process of restating the risks as a set of more detailed risks that will be easier to mitigate, monitor, and manage.**

Risk Mitigation, Monitoring, and Management (RMMM) plan

RMMM Plan : A risk management technique is usually seen in the software Project plan. This can be divided into Risk Mitigation, Monitoring, and Management Plan (RMMM). In this plan, all works are done as part of risk analysis. As part of the overall project plan project manager generally uses this RMMM plan.

In some software teams, risk is documented with the help of a Risk Information Sheet (RIS). This RIS is controlled by using a database system for easier management of information i.e creation, priority ordering, searching, and other analysis. After documentation of RMMM and start of a project, risk mitigation and monitoring steps will start.

Risk Mitigation:

It is an activity used to avoid problems (Risk Avoidance).
Steps for mitigating the risks as follows.

1. Finding out the risk.
2. Removing causes that are the reason for risk creation.
3. Controlling the corresponding documents from time to time.
4. Conducting timely reviews to speed up the work.

Risk Monitoring:

It is an activity used for project tracking.
It has the following primary objectives as follows.

1. To check if predicted risks occur or not.
2. To ensure proper application of risk aversion steps defined for risk.
3. To collect data for future risk analysis.
4. To allocate what problems are caused by which risks throughout the project.

Risk Management and planning :

It assumes that the mitigation activity failed and the risk is a reality. This task is done by Project manager when risk becomes reality and causes severe problems. If the project manager effectively uses project mitigation to remove risks successfully then it is easier to manage the risks. This shows that the response that will be taken for each risk by a manager. The main objective of the risk management plan is the risk register. This risk register describes and focuses on the predicted threats to a software project.

Drawbacks of RMMM:

- It incurs additional project costs.
- It takes additional time.
- For larger projects, implementing an RMMM may itself turn out to be another tedious project.
- RMMM does not guarantee a risk-free project, infact, risks may also come up after the project is delivered

Software Configuration Management

Software Configuration Management (SCM) is a branch of Software Engineering to provide a better process to handling, organizing and controlling the changes in requirements, codes, teams and other elements in the software project development life cycle. The SCM primarily deals with version selection, tracking the changes and version control of software projects with high productivity and minimize the error or risk factor.

Why do we need Software Configuration Management?

It has the following pros to choose as a software configuration management tool in software project development. Such as,

1. Tracking and managing the changes in the software development process.
2. It provides to enhance the Productivity of the software application with minimal error.

3. It provides a smooth workflow inside the development process.
4. Easy to communicate with team members to develop a better quality of the product.
5. To track every member of the team with project workflow status.
6. It updates each team member's code parallelly by considering different version control.
7. To manage the different tools and development processes in a software product.
8. It is used to managing the software and hardware inside the application.
9. It is used to controlling and managing defects, teamwork and process.
10. It handles the software budgeting as per the change in the application.

How does Software Configuration Management work?

A software configuration process is a tool used in the software development life cycle process to track, changing and managing the configuration items (requirements, codes, documents, defects, resources, budgeting, software, hardware, etc) inside the product development. The project manager, developers, configuration manager, the product owner and testers are involved in the SCM process.

It has multiple processes to complete software configuration management.

Such as:

- Planning and Identification Process
- Version Control Process or Baselines
- Change Control Process
- Configuration Release Process
- Configuration Auditing Process
- Review and Status Reporting Process

Advantages

Below are the advantages mentioned:

- To increase the productive efficiency of software as it controls and tracking the workflow or development process.
- It welcomes change management so the risk of the product will be less.

- It is used for proper monitoring and auditing of the software development product.
- It will help to enhance the software development life cycle process.
- This process provides a reliable, organized, cost-effective and low-risk software development application.
- It provides a high-quality software product.

Disadvantages

Below are the disadvantages:

- It needs adequate resources with full knowledge about the software configuration management tools.
- It requires more resources to work with the configuration management process for small industries.
- It requires a highly configured desktop/laptop for the development stages.

What is SCM repository?

Source code management (SCM) is used to track modifications to a source code repository. SCM tracks a running history of changes to a code base and helps resolve conflicts when merging updates from multiple contributors.

What is software configuration management repository?

Software configuration management is referred to as source control, change management, and version control. ... Any software configuration management system creates a central repository to facilitate file sharing. Each file to be shared must be added to the central repository to create the first version of the file.

What are the tasks of SCM process?

The five tasks of the SCM process are **configuration identification, change control, version control, and configuration auditing, and reporting.**

1. **Configuration identification** is at the core of the SCM process. It sets the basis for the subsequent tasks of the SCM process. Configuration identification includes defining the basis for identifying software configuration items (SCIs), defining the method to depict

the relationship between SCIs, defining the identification scheme for naming SCIs, identifying SCIs, identifying the baselines to be established and the corresponding SCIs, and defining the method for acquiring SCIs in the project repository.

2. In the **change control task** of the SCM process, a defined process is used to handle change requests from their initiation until the time the change is implemented and released to users. The change control process includes evaluating a change request, implementing the change, and verifying and releasing the change.
3. The **version control task** of the SCM process is executed to manage multiple versions of a configuration and the SCIs that constitute it. Revisions, variations, and variants are types of versions of a software product; and the principles of version control apply to all of them.
4. The **configuration auditing task** of the SCM process is a software quality assurance (SQA) function. It is executed formally and impartially to ensure that changes have been made correctly and quality has been maintained while implementing the changes. There are three types of configuration audits—functional configuration audit (FCA), physical configuration audit (PCA), and formal qualification review (FQR).
5. The **reporting task** of the SCM process provides timely information on the status of the changes requested and the SCIs to the people who may be affected by the changes. These could be the people requesting for changes, the developers, the project manager, and the senior management.

What are the five stages of the configuration management process?

Here's a refresher on how to do it.

- First, Plan. Configuration management begins at the beginning of the project when you're putting together your project management plan. ...
- Next, Identify Config Items. ...
- Control Your Records. ...
- Status Accounting. ...
- Finally, Audit.

Risk Management Activities



Risk Assessment

Software risk assessment is a **process of identifying, analyzing, and prioritizing risks**

The objective of risk assessment is to division the risks in the condition of their loss, causing potential. For risk assessment, first, every risk should be rated in two methods:

- The possibility of a risk coming true (denoted as r).
- The consequence of the issues relates to that risk (denoted as s).

Based on these two methods, the priority of each risk can be estimated:

$$p = r * s$$

Where p is the priority with which the risk must be controlled, r is the probability of the risk becoming true, and s is the severity of loss caused due to the risk becoming true. If all identified risks are set up, then the most likely and damaging risks can be controlled first, and more comprehensive risk abatement methods can be designed for these risks.

1. **Risk Identification:** The project organizer needs to anticipate the risk in the project as early as possible so that the impact of risk can be reduced by making effective risk management planning.

2. **Risk Analysis** : In this stage or process, the likelihood and consequences of these risks assessed.

3. **Risk Prioritization**: In the risk prioritization step, the overall set of identified risk events, their impact assessments, and their probabilities of occurrences are "processed" to derive a most-to-least-critical rank-order of identified risk.

Risk Control

It is the process of managing risks to achieve desired outcomes.

There are three main methods to plan for risk management:

1. **Avoid the risk**: This may take several ways such as discussing with the client to change the requirements to decrease the scope of the work, giving incentives to the engineers to avoid the risk of human resources turnover, etc.
2. **Transfer the risk**: This method involves getting the risky element developed by a third party, buying insurance cover, etc.
3. **Risk reduction**: This means planning method to include the loss due to risk. For instance, if there is a risk that some key personnel might leave, new recruitment can be planned.

1.Risk planning: The risk planning method considers each of the key risks that have been identified and develop ways to maintain these risks.

2.Risk Monitoring: Risk monitoring is the method king that your assumption about the product, process, and business risks has not changed.

3.Risk Resolution : Risk resolution produces a situation in which the risk items are eliminated or otherwise resolved